EXPERIMENT

Data Interpretation – PPO, TRPO & DDPG Performance Analysis

Evaluate – Level 5

Objective

Implement PPO, TRPO and DDPG-based reinforcement learning experiments and evaluate their performance using reward, learning stability and convergence behaviour.

Problem Statement

Compare policy-gradient and actor-critic algorithms on continuous-control tasks. Interpret training rewards and identify which method provides better learning performance under the same experimental setting.

Required Analysis

• Implement PPO and DDPG with PyTorch.

• Implement a practical TRPO-style constrained policy update.

• Train the algorithms on a continuous-control environment.

• Record episode rewards and moving-average performance.

• Compare final reward, best reward and learning stability.

• Interpret the performance differences from the generated results.

Algorithms

PPO (Proximal Policy Optimization): limits policy changes using a clipped objective.

TRPO (Trust Region Policy Optimization): constrains the policy update using a KL-divergence trust region.

DDPG (Deep Deterministic Policy Gradient): uses an actor and critic with deterministic actions, replay memory and target networks.

1. Performance Measures and Data Interpretation

Episode Reward

The total reward obtained in an episode is the primary performance measure. Higher reward generally indicates that the agent is performing the task more successfully.

Moving Average

A moving average smooths noisy episode rewards and makes the learning trend easier to interpret. A rising moving average indicates improvement, while a flat curve indicates limited learning.

Final Average Reward

The mean reward over the final set of episodes is used to estimate the performance of the trained agent after learning has progressed.

Best Reward

The maximum episode reward indicates the highest performance reached during training. It should be considered together with the average because one unusually high episode does not necessarily indicate stable learning.

Stability

An algorithm is considered more stable when its reward curve improves with relatively smaller fluctuations and maintains a higher moving average.

Interpretation Rules

• Higher final average → stronger final performance.

• Higher best reward → better peak performance.

• Smoother increasing curve → better learning stability.

• Early rapid improvement → faster learning.

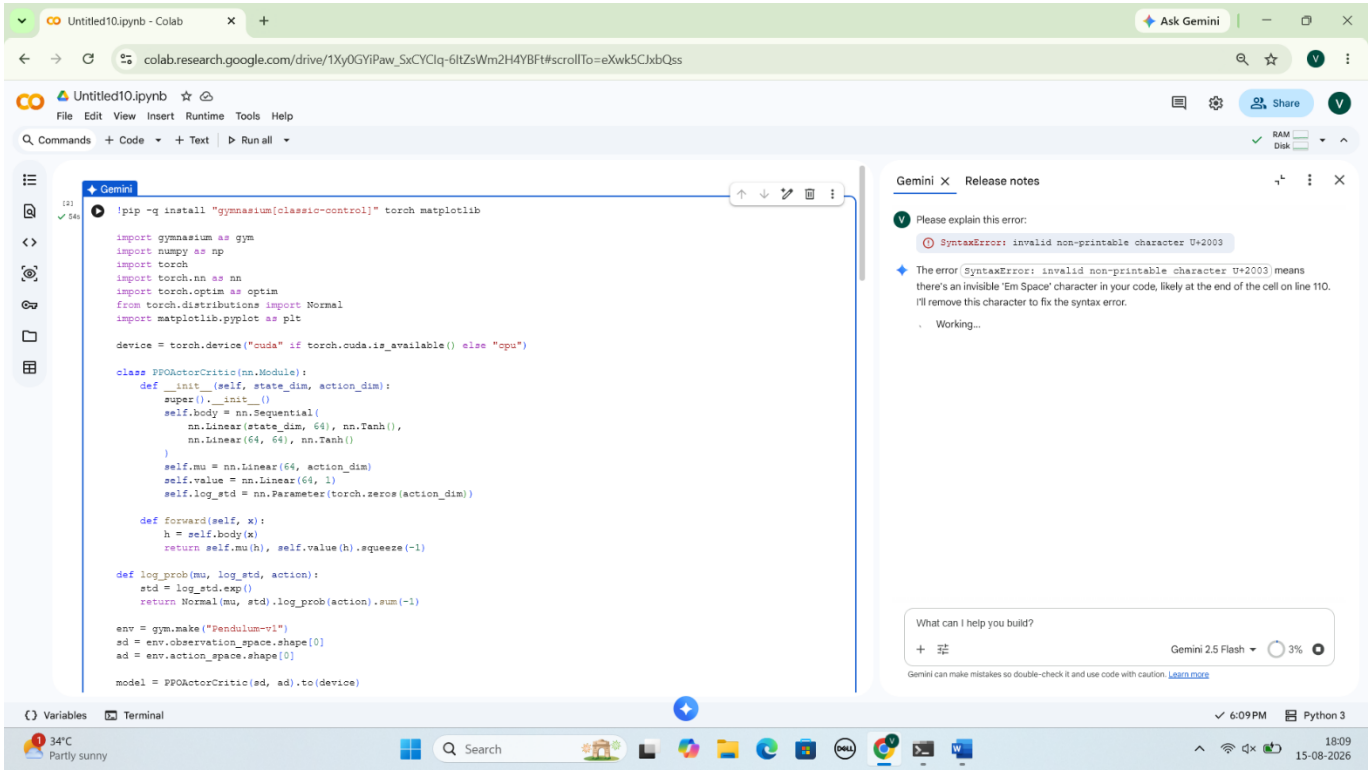
• Large reward fluctuations → higher training variance.

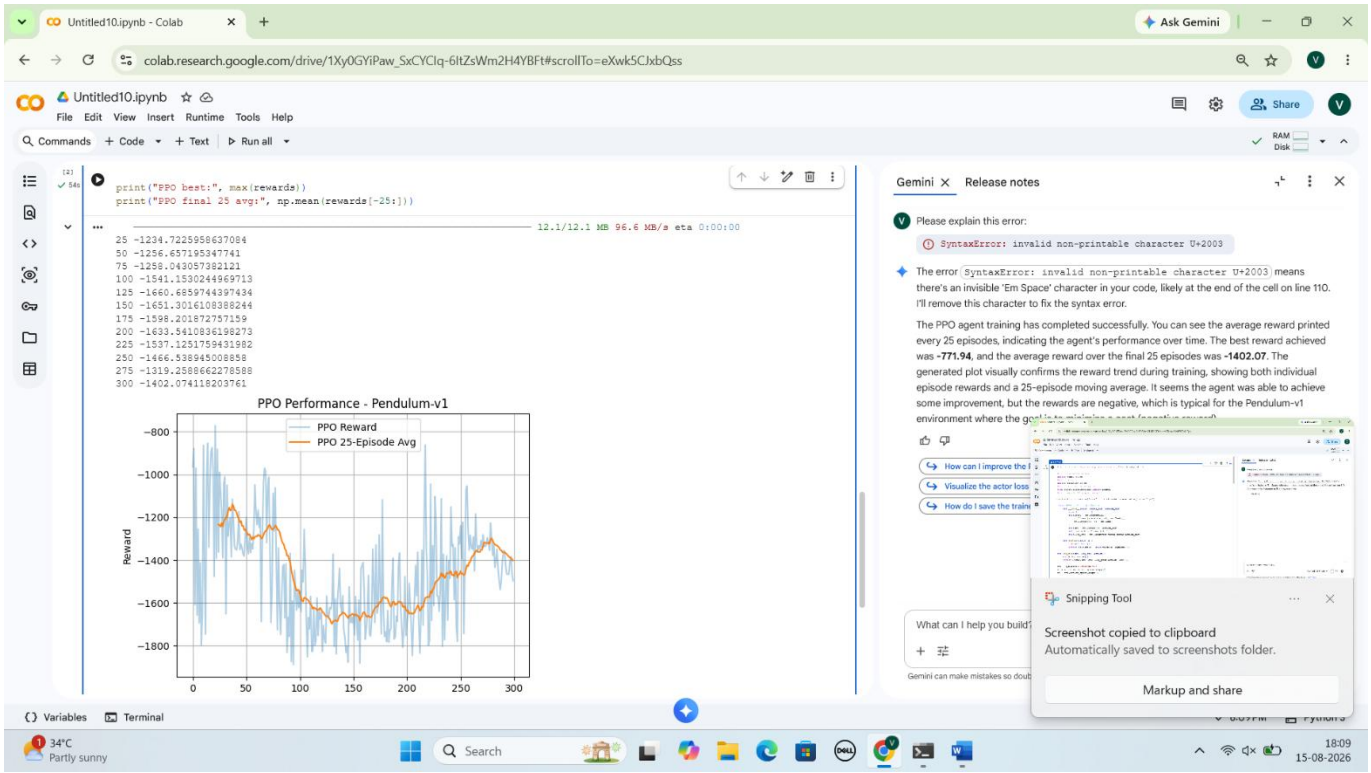
Expected Comparison

PPO is generally expected to provide stable learning because its clipped objective limits excessively large policy updates. TRPO explicitly controls policy changes through a trust-region constraint. DDPG can perform well in continuous action spaces but may be more sensitive to exploration, hyperparameters and replay-buffer quality.

2. Google Colab Implementation – PPO

The following code is self-contained and uses PyTorch. It trains PPO on Pendulum-v1, a continuous-control environment, and prints performance statistics.

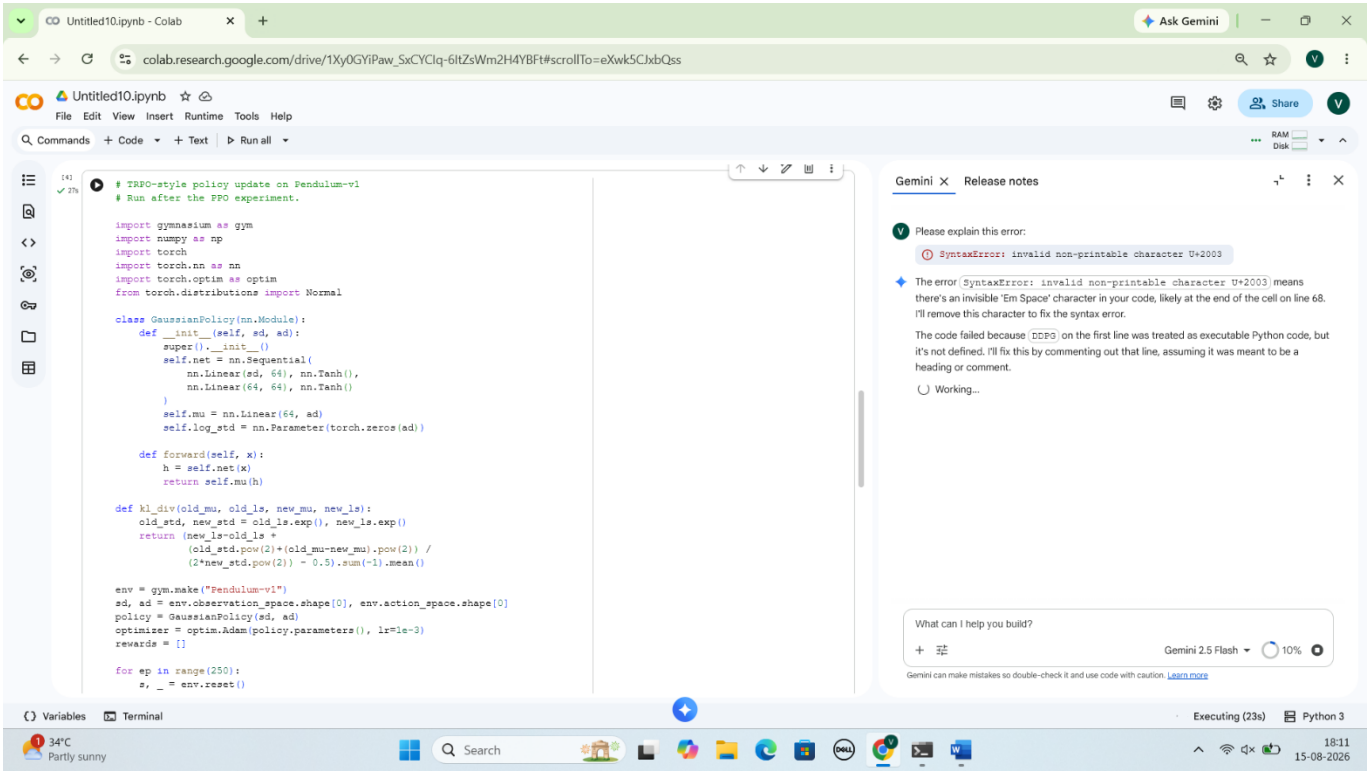




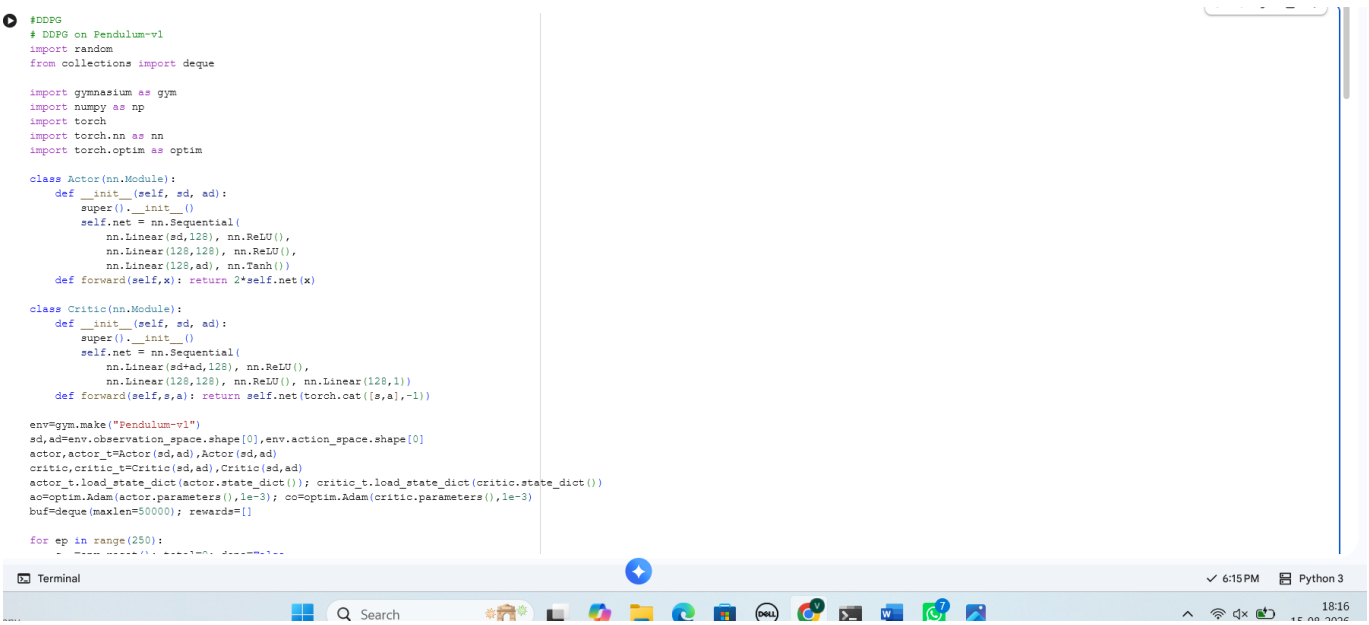
3. Google Colab Implementation – TRPO-style and DDPG

TRPO-style update

This compact implementation demonstrates the central TRPO idea: reject a proposed policy update when the KL divergence exceeds a trust-region threshold. It is designed for clarity and evaluation rather than reproducing every optimization detail of the original TRPO paper.



DDPG



4. Performance Evaluation and Data Interpretation

Run the three experiments and record the printed values in the table below.

Algorithm	Best Reward	Final 25-Episode Average	Stability / Observation
PPO	Enter Colab result	Enter Colab result	Interpret reward curve
TRPO	Enter Colab result	Enter Colab result	Interpret reward curve
DDPG	Enter Colab result	Enter Colab result	Interpret reward curve

Data Interpretation

1. Compare the final 25-episode average. The algorithm with the highest value demonstrates the strongest final performance in this experiment.

2. Compare the best reward to determine which algorithm reached the highest peak performance.

3. Inspect the moving-average curve. A smoother curve with fewer large drops indicates better training stability.

4. If PPO reaches a high final average with moderate fluctuations, interpret it as stable policy optimization.

5. If TRPO maintains smaller policy changes and a smooth learning trend, relate this to its trust-region constraint.

6. If DDPG improves quickly but fluctuates strongly, relate the behaviour to exploration noise, replay-buffer sampling and sensitivity to hyperparameters.

7. Do not claim that one algorithm is universally best. The conclusion must be based on the actual Colab results obtained under the same environment, training budget and evaluation criteria.

Evaluation Criteria

• Reward performance: final average and best reward.

• Learning stability: smoothness and consistency of the moving average.

• Convergence: whether performance reaches a stable high level.

• Sample efficiency: how quickly reward improves during training.

• Algorithm behaviour: explain results using the main mechanism of PPO, TRPO or DDPG.